

Dynamics of an Insulated Rail Joint

Principles of Machine Design with FEM Applications
Exam project

Candidate: Lorenzo Dalla Rosa

Supervisors: Ass. Prof. Ing. Giancarlo Santamato

Prof. Ing. Massimiliano Solazzi

Prof. Ing. Antonio Frisoli



Sant'Anna
Scuola Universitaria Superiore Pisa

| **Istituto di Intelligenza Meccanica**

Project description: Insulated Rail Joint

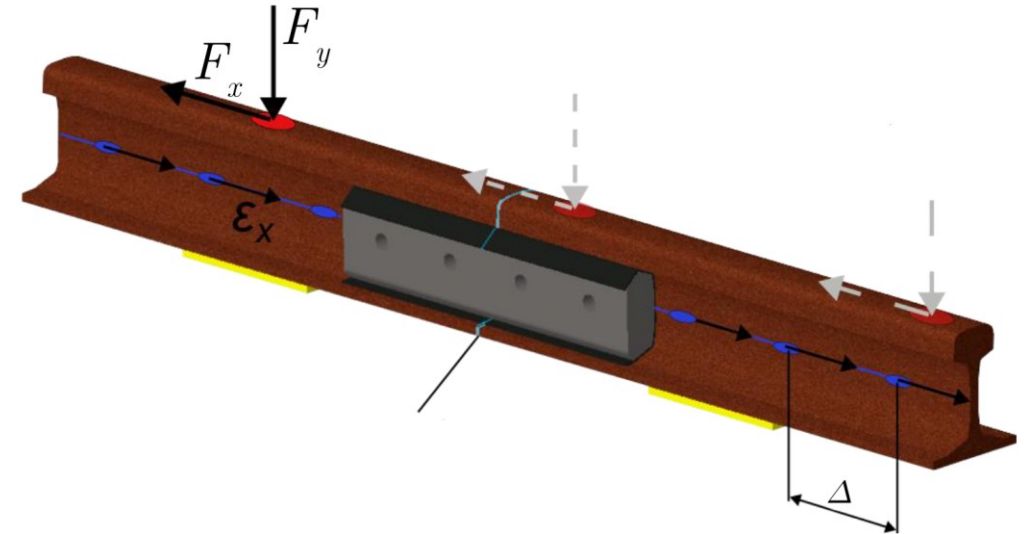
Function: Ensures structural continuity of Continuous welded rails maintaining electrical isolation for signalling circuits

Components:

- Rails (separated by a standard gap)
- Fishplates with insulating liners
- Insulating End-Post: focus of the failure simulation

Boundary conditions:

- Hertzian train load travelling at $v = 8$ m/s
($F_z = 150$ kN, $F_x = 30$ kN)



$$p(x, y) = p_0 \sqrt{1 - \frac{x^2}{a^2} - \frac{y^2}{b^2}} \quad p_0 = \frac{3F_z}{2\pi ab}$$

- Elastic support

Direction	Symbol	Coefficient [MN/m]
Longitudinal (x)	k_x	10
Transverse (y)	k_y	15
Vertical (z)	k_z	100

Probes:

- Spatial resolution: at least 100 mm
- Longitudinal strain ϵ_x

Task:

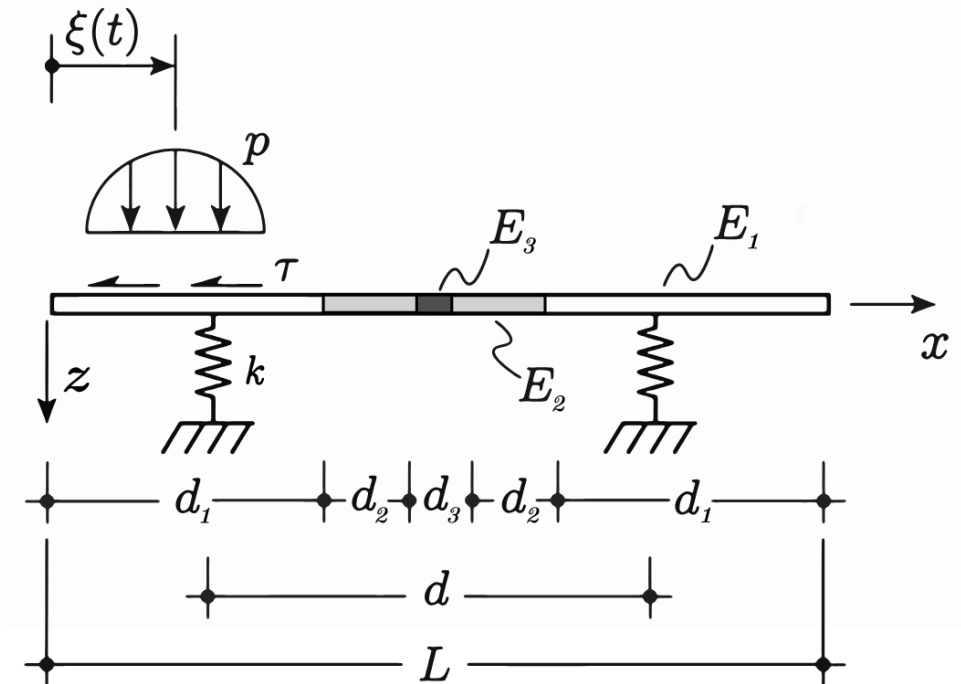
- Reconstruct the strain wave $\epsilon_x(x, t)$
- Simulate failure of insulating end post

Analytical model

$$\begin{cases} m(x) \frac{\partial^2 w(x, t)}{\partial t^2} + \frac{\partial^2}{\partial x^2} \left[EI(x) \frac{\partial^2 w(x, t)}{\partial x^2} \right] = p(x - vt) - k_z(x) w(x, t) \\ m(x) \frac{\partial^2 u(x, t)}{\partial t^2} - \frac{\partial}{\partial x} \left[EA(x) \frac{\partial u(x, t)}{\partial x} \right] = \tau(x - vt) - k_x(x) u(x, t) \\ + \text{B.C.} \end{cases}$$

$$\begin{cases} w(x_i^-) = w(x_i^+) & (\text{Continuity of displacement}) \\ \frac{\partial w}{\partial x}(x_i^-) = \frac{\partial w}{\partial x}(x_i^+) & (\text{Continuity of slope}) \\ EI(x_i^-) \frac{\partial^2 w}{\partial x^2}(x_i^-) = EI(x_i^+) \frac{\partial^2 w}{\partial x^2}(x_i^+) & (\text{Equilibrium of bending moment}) \\ \frac{\partial}{\partial x} \left(EI \frac{\partial^2 w}{\partial x^2} \right)_{x_i^-} = \frac{\partial}{\partial x} \left(EI_2 \frac{\partial^2 w}{\partial x^2} \right)_{x_i^+} & (\text{Equilibrium of shear force}) \end{cases}$$

$$\begin{cases} u(x_i^-) = u(x_i^+) & (\text{Continuity of axial displacement}) \\ EA(x_i^-) \frac{\partial u}{\partial x}(x_i^-) = EA(x_i^+) \frac{\partial u}{\partial x}(x_i^+) & (\text{Equilibrium of axial force}) \end{cases}$$



The problem:

The rail is a piecewise continuous structure. Solving the non-homogeneous equations independently across three distinct zones with varying properties (EI, EA, m) is exceptionally cumbersome

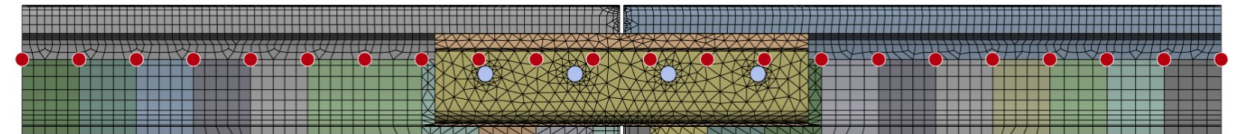
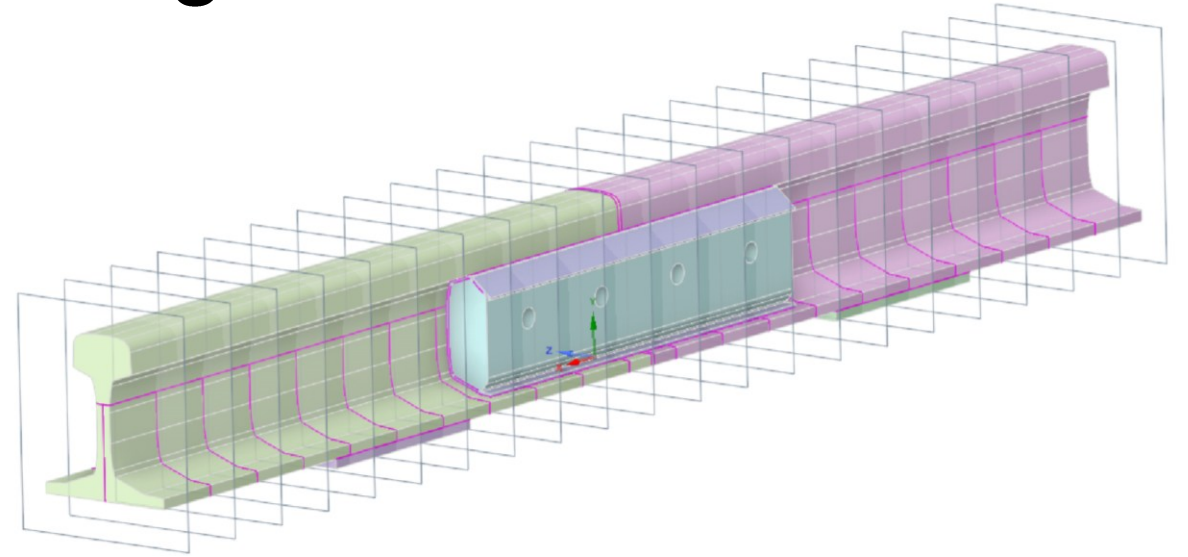
The solution:

Finite element method

Geometry partitioning and meshing

Geometry preparation in Ansys SpaceClaim

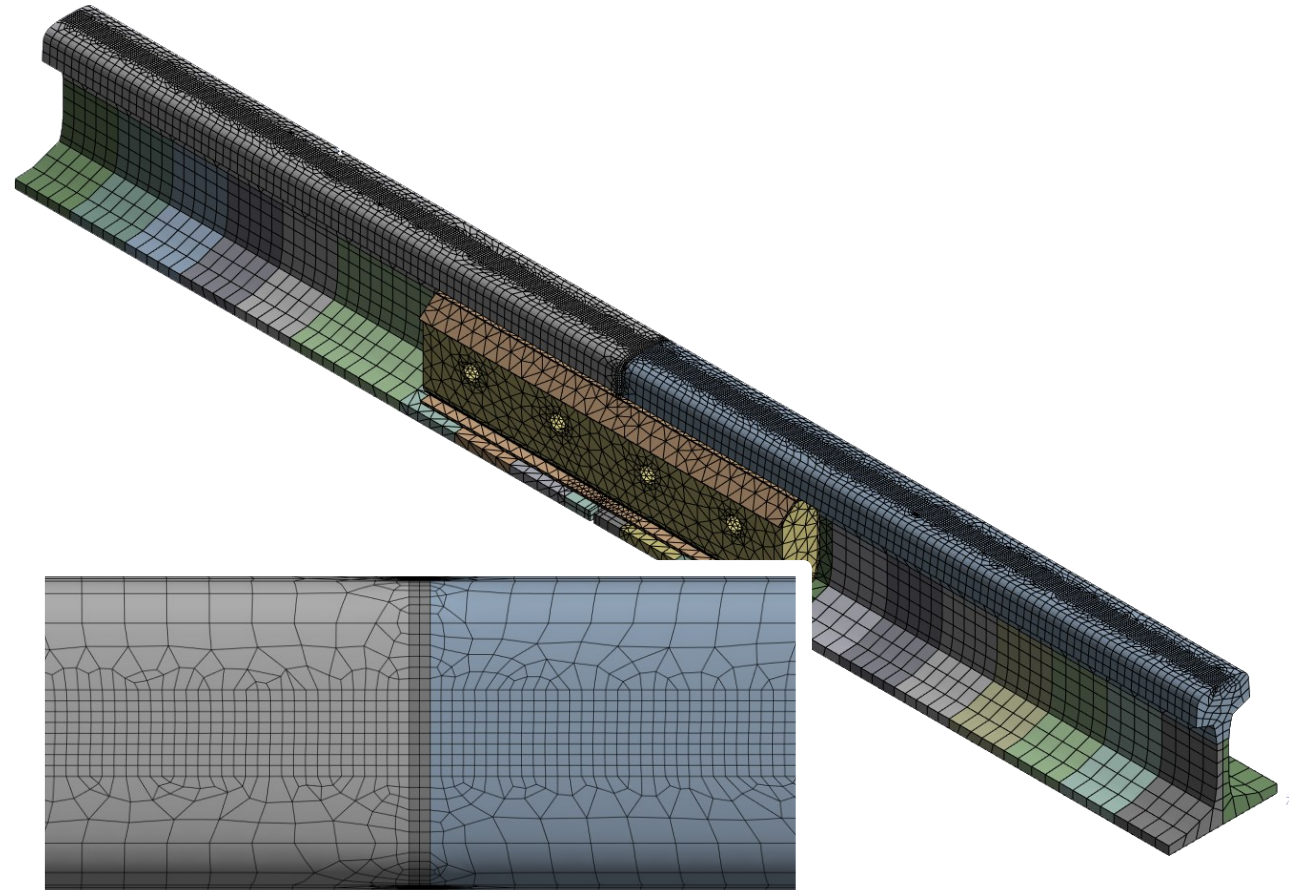
- Partitioned the model into **multiple bodies** to force exact **node alignment** with the **22** probes (**76.43 mm** interval)
- Separated prismatic sections to generate to force the mesher to produce higher quality elements
- Split faces that will become the elastic support



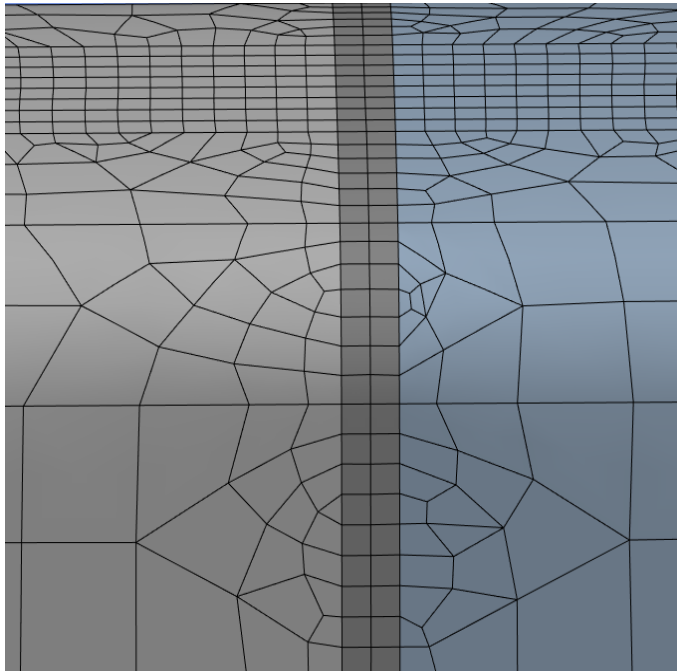
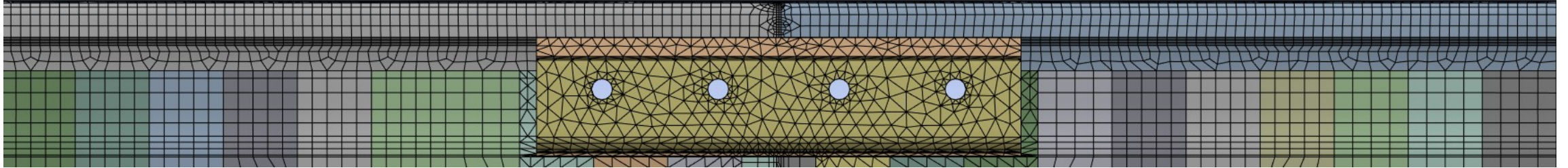
Geometry partitioning and meshing

Meshing in Ansys Mechanical Workbench

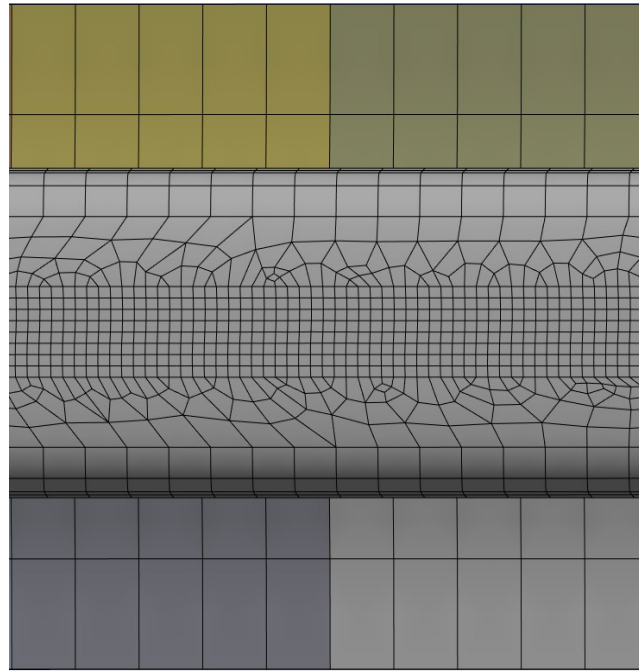
- **Hex-dominant** method for upper rail and end-post
- **Tetrahedra** mesh for fish-plate
- **Local Refinement:** **2.5 mm** element size
 - at the **load application zone** to model accurately the **10 mm** minor axis of the Hertzian contact ellipse
 - at the **end-post** to avoid **artificial stiffness**
 - Additional local refinements for **bolt holes**
- **Optimization:** Coarse **16 mm** Swept mesh on the rail base to reduce element count
- **Iterative process:** balancing fidelity over computational resources



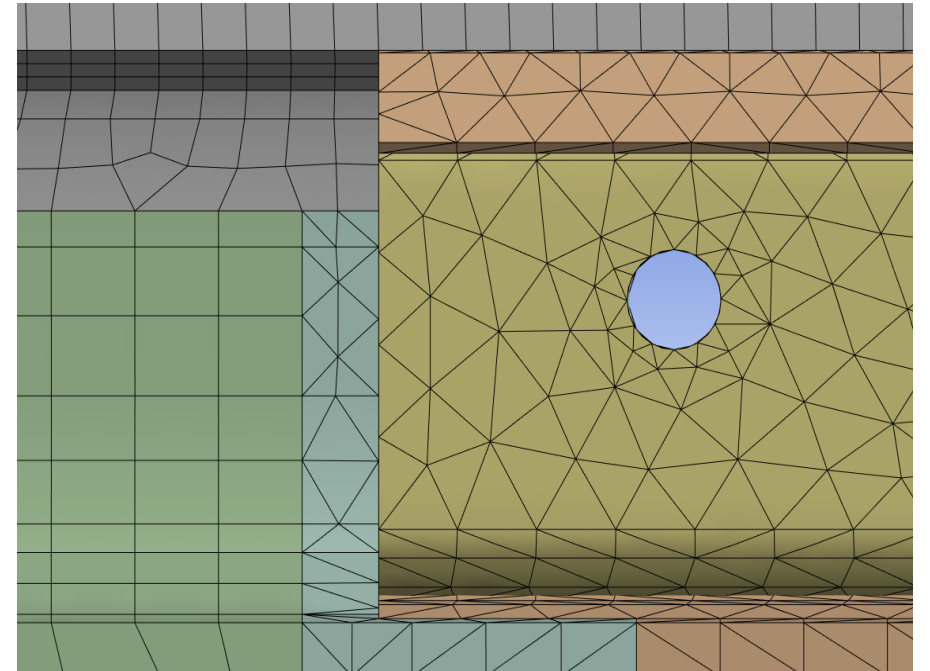
Meshing



End-post



Load application patch



Fishplate and hole treatment

APDL Modelling

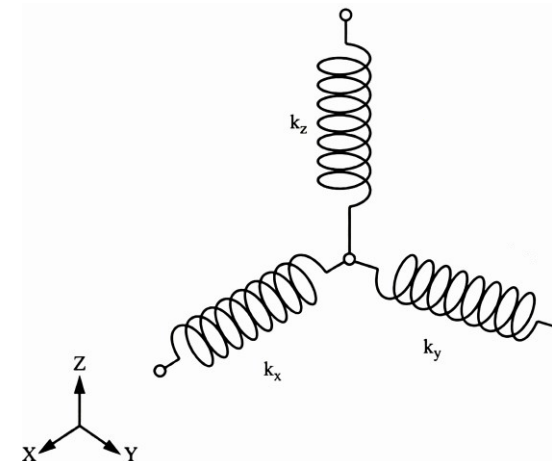
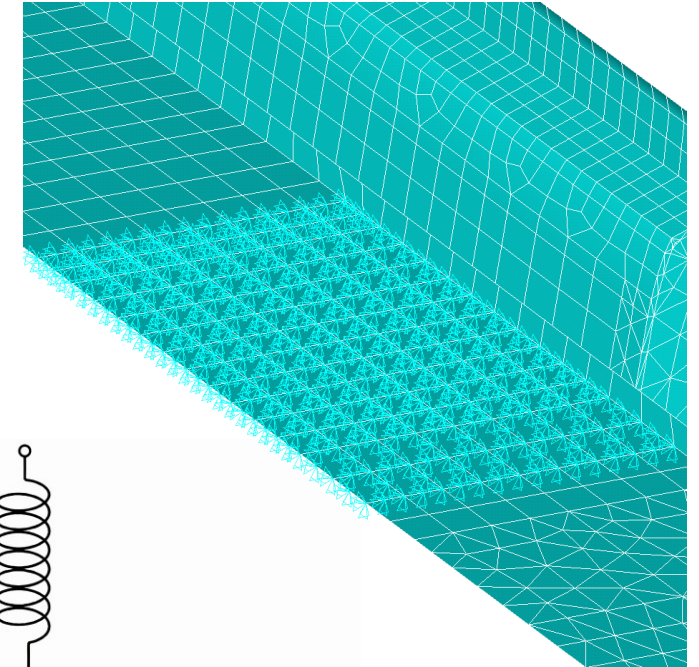
ANSYS Mechanical Workbench does not support surface loads varying simultaneously in space and in time:

$$p = p(x, y, t)$$

Elastic Foundation: COMBIN14 spring elements to simulated distributed nodal stiffness

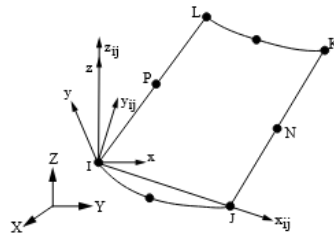
$$k_{\text{node},i} = \frac{k_i}{N_{\text{nodes}}} \quad i = x, y, z$$

Direction	Symbol	Coefficient [MN/m]
Longitudinal (x)	k_x	10
Transverse (y)	k_y	15
Vertical (z)	k_z	100

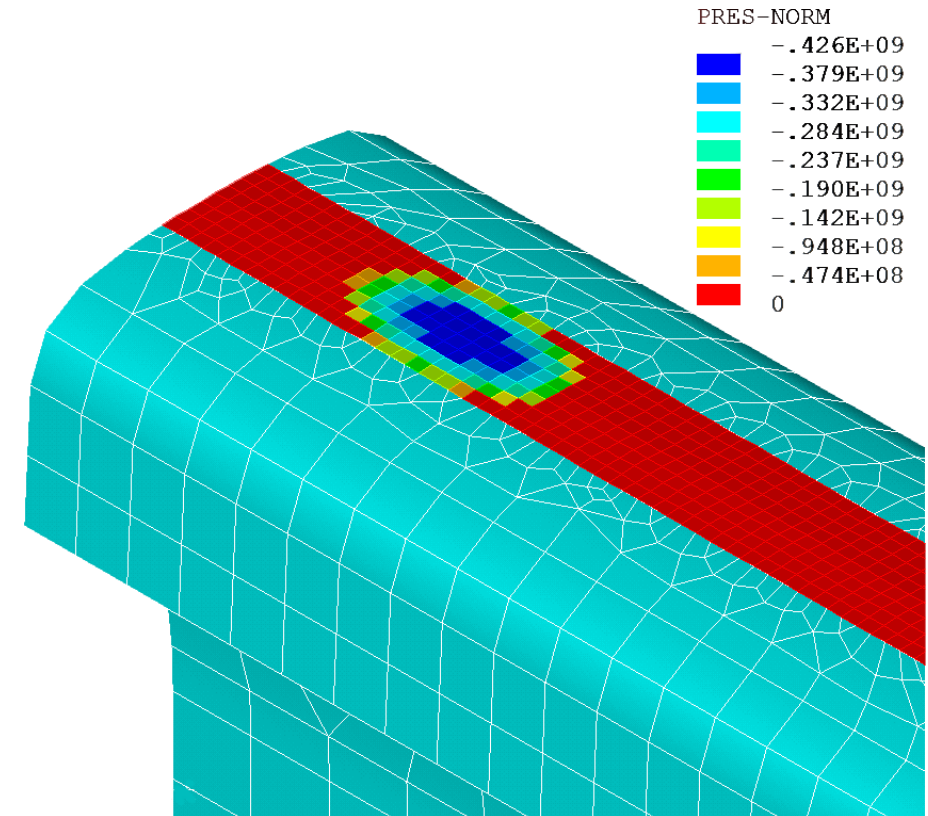
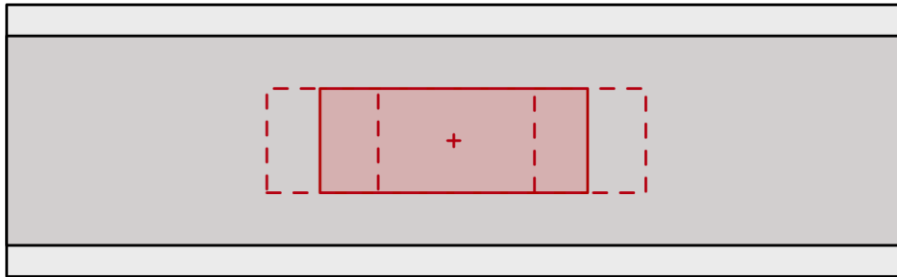


APDL Modelling

Moving load application: SURF154 surface effect elements generated on the rail head



Spatial Pre-filtering: instead of calculating null pressures outside of the contact zones, the code loops over a restricted bounding box cantered on the load, drastically reducing computation



Transient analysis

(1) Linear analysis

- **Low strain** peaks
- $\sigma < \sigma^Y$
- Linear constitutive law $\{\sigma\} = [C]\{\varepsilon\}$

(2) Time step

- Fixed time step $\Delta t = 0.2 \text{ ms} \Rightarrow f_N = 2500 \text{ Hz}$
- The pressure print falls **25 times** within the contact ellipse reducing **the introduction of artifacts**

(3) Preconditioned Conjugate Gradient solver vs Direct Solver

- **PCG**: iterative approach to converge on a solution, making it significantly more memory-efficient for models with many degrees of freedom
- **Sparse Direct Solver**: instead of approximating the solution through successive iterations, the solver factorizes the stiffness matrix $[K]$ into lower and upper triangular matrices

Transient analysis

(4) HHT time integration method

- Suppresses un-physical high-frequency numerical noise
- **Energy dissipation method**

$$D_{\text{ratio}} = \frac{E_{\text{diss}}}{E_{\text{mech}}} = \frac{|E_{\text{mech}} - W_{\text{ext}}|}{E_{\text{mech}}}$$

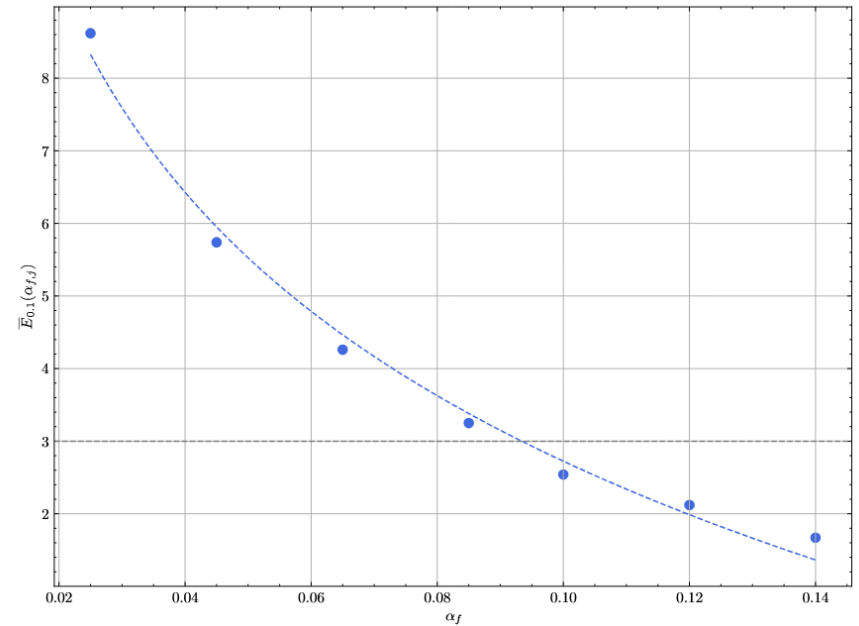
- **Parametric study**

$$[M]\{\ddot{u}(t)\} + [K]\{u(t)\} = \{F^a(t)\}$$

$$[M]\{\ddot{u}_{n+1}\} + (1 - \alpha_f)[K]\{u_{n+1}\} + \alpha_f[K]\{u_n\} = (1 - \alpha_f)[F_{n+1}] + \alpha_f[F_n]$$

$$\{u_{n+1}\} = \{u_n\} + \Delta t\{\dot{u}_n\} + \Delta t^2 \left[\left(\frac{1}{2} - \beta \right) \{\ddot{u}_n\} + \beta\{\ddot{u}_{n+1}\} \right]$$

$$\{\dot{u}_{n+1}\} = \{\dot{u}_n\} + \Delta t[(1 - \gamma)\{\ddot{u}\}_n + \gamma\{\ddot{u}_{n+1}\}]$$

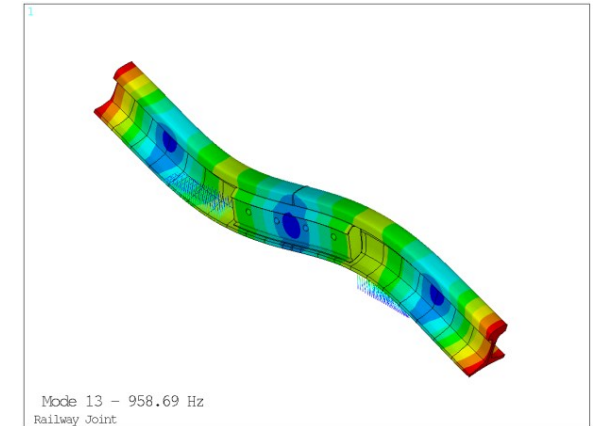
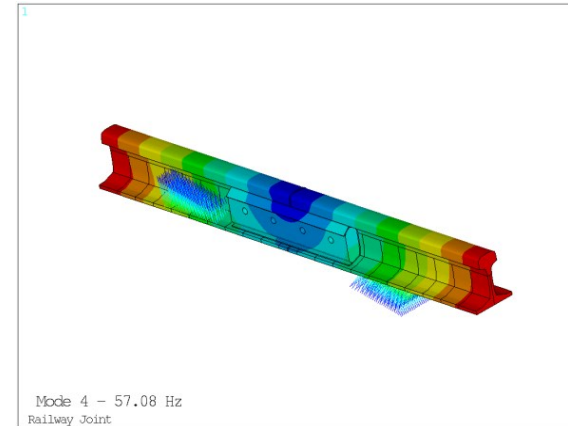


$$\gamma = \frac{1}{2} + \alpha_f \quad \text{and} \quad \beta = \frac{(1 + \alpha_f^2)}{4}$$

Modal analysis

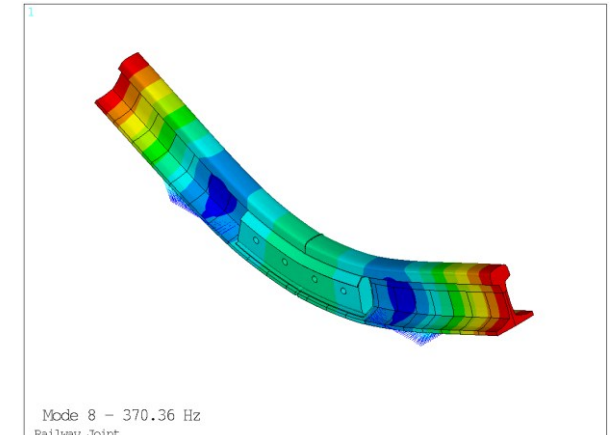
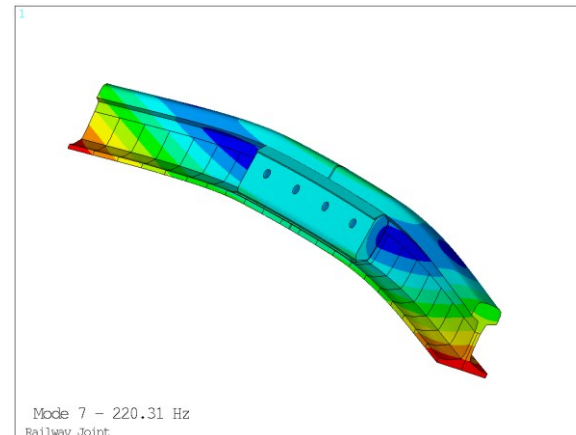
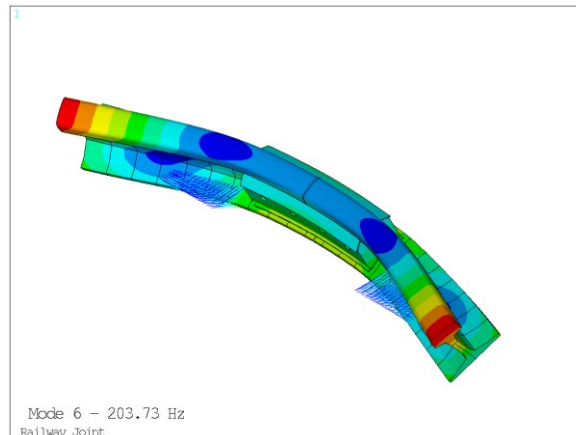
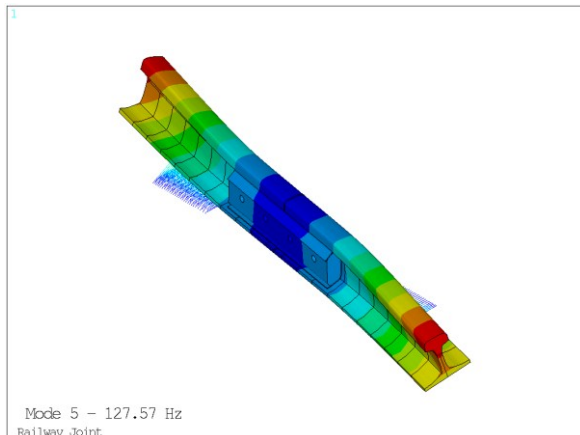
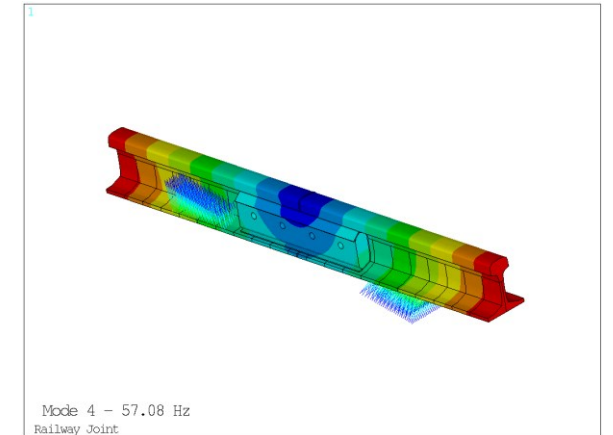
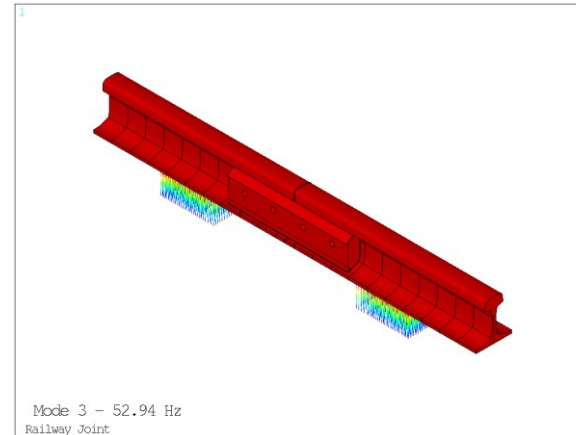
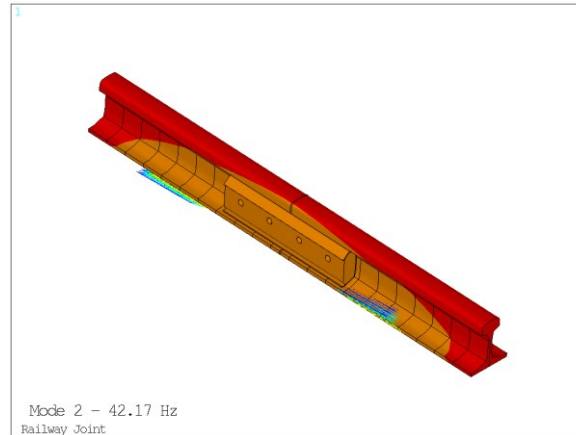
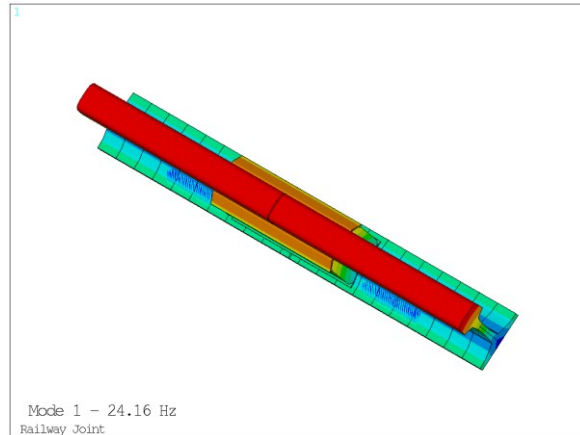
$$[M]\{\ddot{u}\} + [K]\{u\} = \{0\} \text{ Assuming harmonic motion: } ([K] - \omega_i^2[M])\{\varphi_i\} = \{0\}$$

- Extracted the first **25 natural frequencies** and **associated mode shapes**
- **Nyquist Limit:** frequencies above **2.5 kHz** were ignored due to the sampling rate
- **Semi-rigid mode shapes:** up to the 4th mode
- **Local Modes:** subsequent natural frequencies are associated to local modes
- **Mesh independence:** sub 1Hz variation in natural frequency
- **Damage effect:** no significant variation in natural frequencies is introduced by the damaged end-post

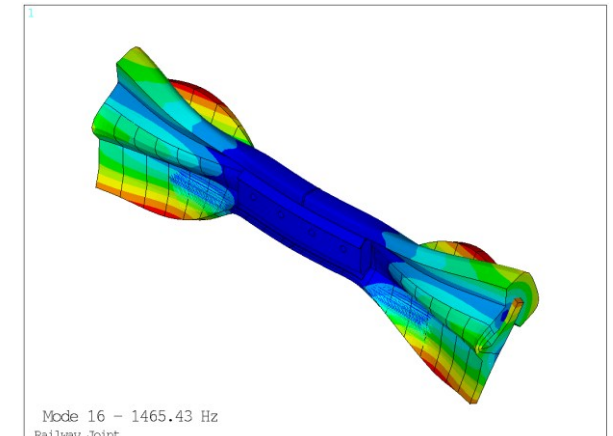
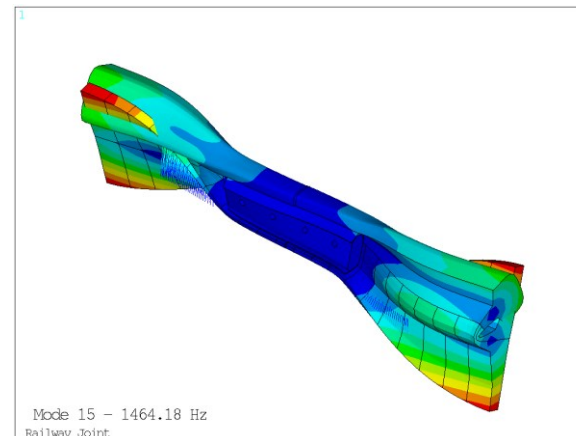
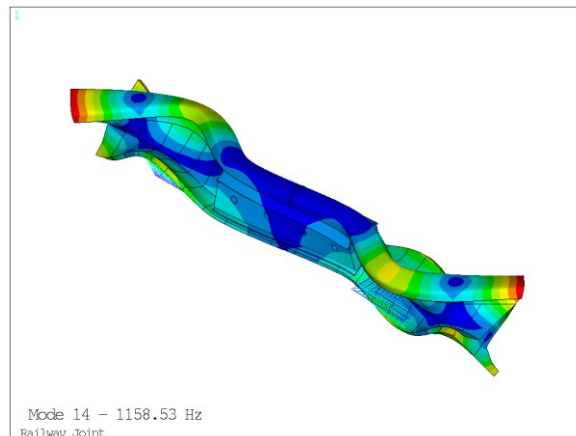
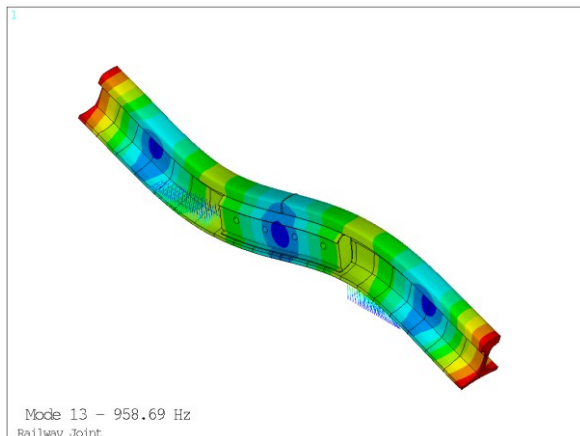
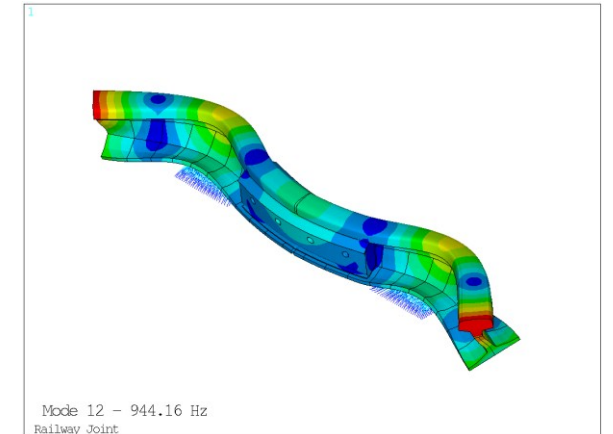
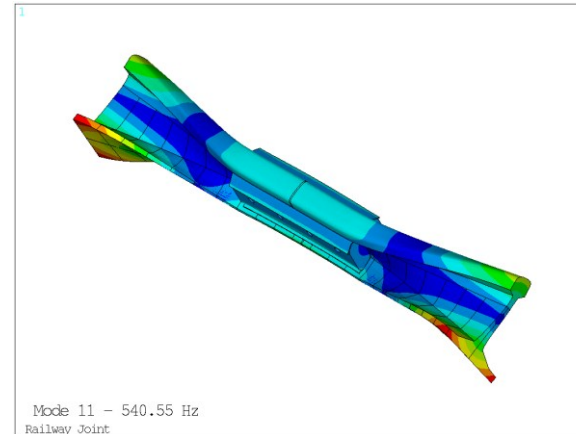
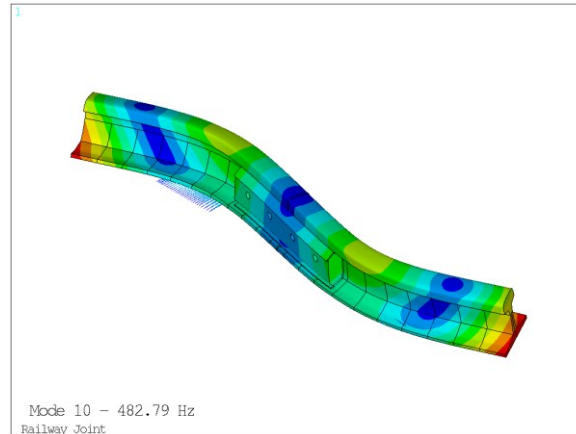
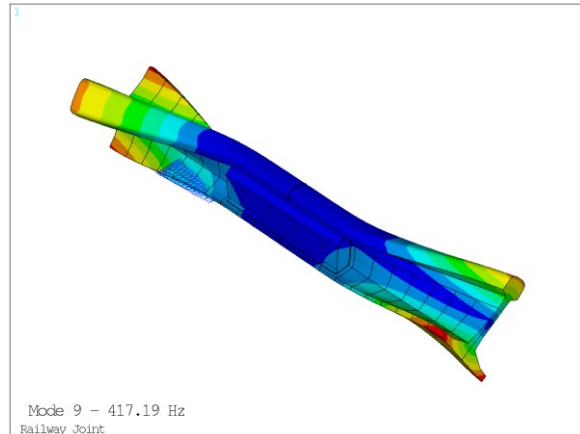


Mode	f [Hz]	x		y		z		ROT _x		ROT _y		ROT _z	
		Γ _x	Σm _x	Γ _y	Σm _y	Γ _z	Σm _z	Γ _{θx}	Σm _{θx}	Γ _{θy}	Σm _{θy}	Γ _{θz}	Σm _{θz}
1	24.17	0.00	0.00	0.00	0.00	9.17	0.62	1.18	1.00	0.00	0.00	0.00	0.00
2	42.18	11.22	0.93	0.00	0.00	0.00	0.62	0.00	1.00	0.00	0.00	-2.10	0.19
3	52.93	0.00	0.93	11.61	1.00	0.00	0.62	0.00	1.00	0.00	0.00	0.00	0.19
4	57.08	3.00	1.00	0.00	1.00	0.00	0.62	0.00	1.00	0.00	0.00	4.29	1.00
5	127.54	0.00	1.00	0.00	1.00	0.00	0.62	0.00	1.00	4.60	0.98	0.00	1.00
6	202.48	0.00	1.00	0.00	1.00	-4.08	0.75	0.01	1.00	0.00	0.98	0.00	1.00
7	220.15	0.00	1.00	0.00	1.00	5.83	1.00	-0.01	1.00	0.00	0.98	0.00	1.00
8	333.55	0.00	1.00	0.08	1.00	0.00	1.00	0.00	1.00	0.00	0.98	0.00	1.00
9	417.02	0.00	1.00	0.00	1.00	0.00	1.00	0.00	1.00	-0.65	1.00	0.00	1.00
10	482.43	0.00	1.00	0.00	1.00	0.00	1.00	0.00	1.00	-0.24	1.00	0.00	1.00
11	540.38	0.00	1.00	0.00	1.00	0.06	1.00	0.00	1.00	0.00	1.00	0.00	1.00
12	937.65	0.00	1.00	0.00	1.00	-0.21	1.00	0.00	1.00	0.00	1.00	0.00	1.00
13	958.17	-0.01	1.00	0.00	1.00	0.00	1.00	0.00	1.00	0.00	1.00	0.01	1.00

Mode shapes



Mode shapes



Data analysis

Raw data extraction

- Load step
- Extract x component of **EPEL** in global coordinate system
- Save to **.csv**



Cubic spline interpolation

- Data interpolation in Python
- Wave reconstruction
- **Physical justification:** a cubic spline minimizes the integral of the second spatial derivative

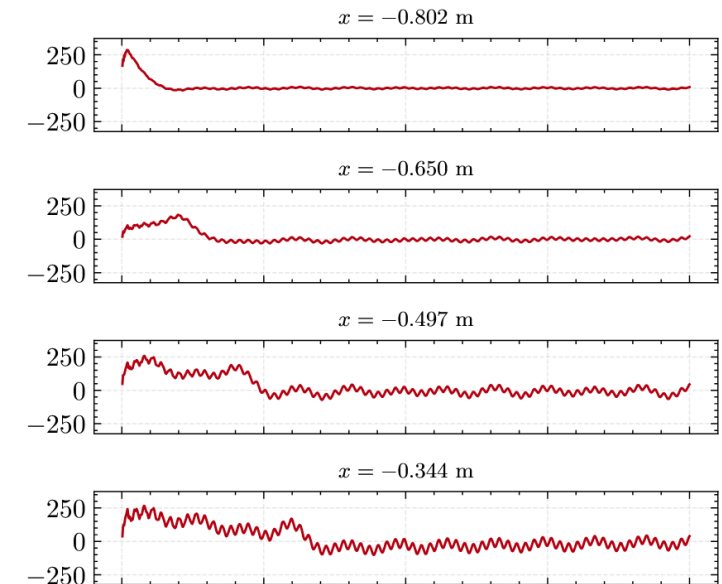
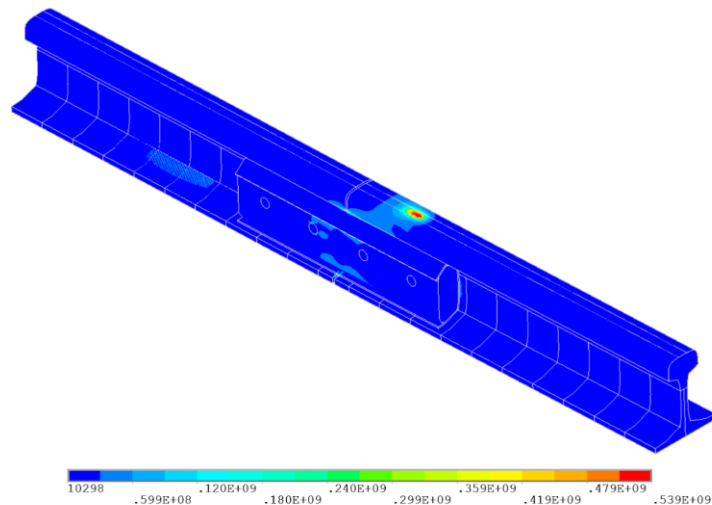
$$\min \int_0^L [S''(x)]^2 dx$$

$$U = \frac{1}{2} \int_0^L EI \left(\frac{\partial^2 v}{\partial x^2} \right)^2 dx$$



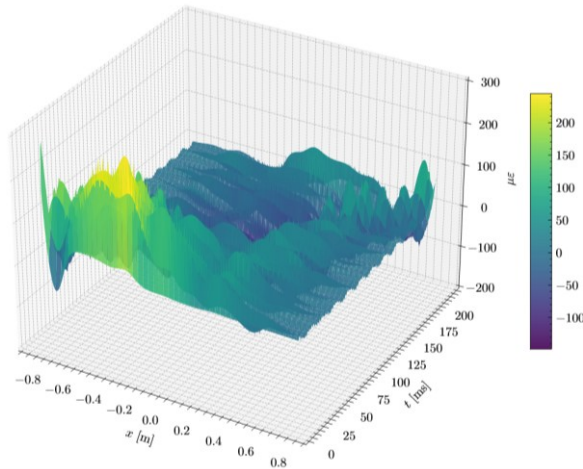
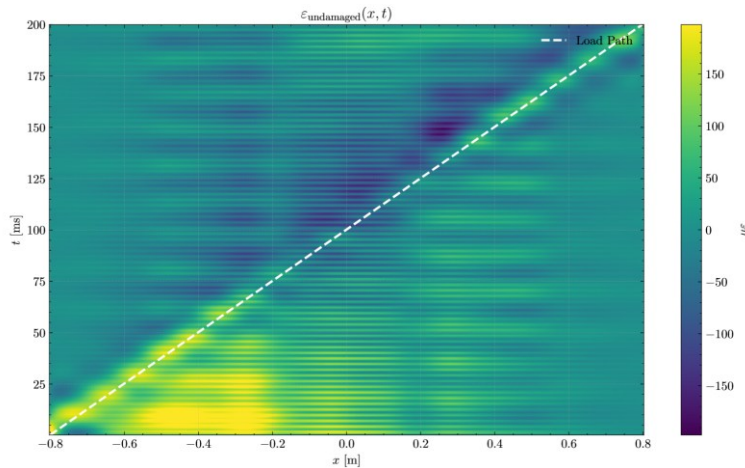
Spectral density analysis

- Using **Welch's method** we can **compute the Power spectral density**
- Both **per probe spectra** and **average spectrum**

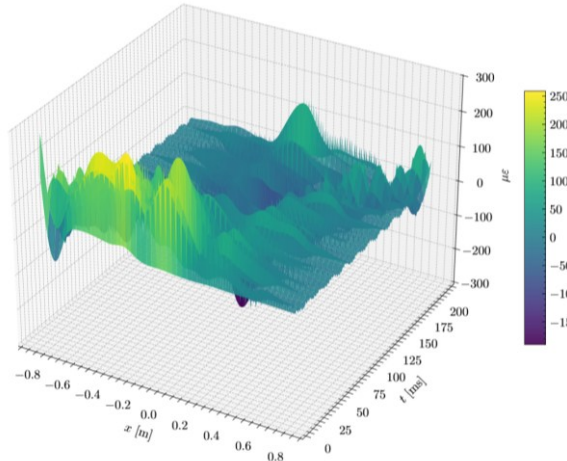
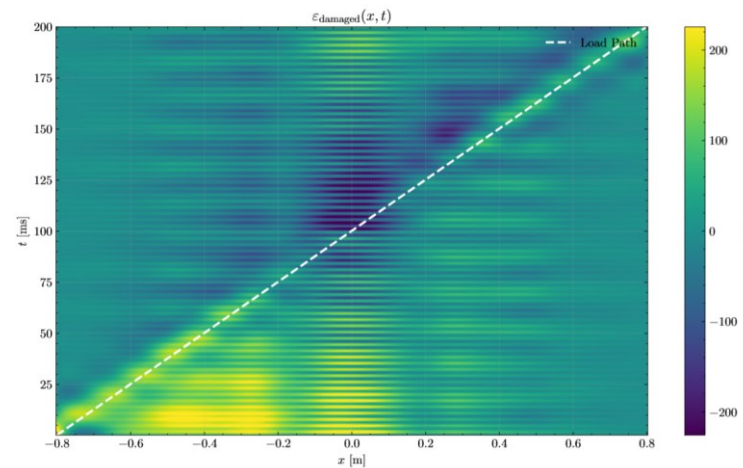


Wave reconstruction

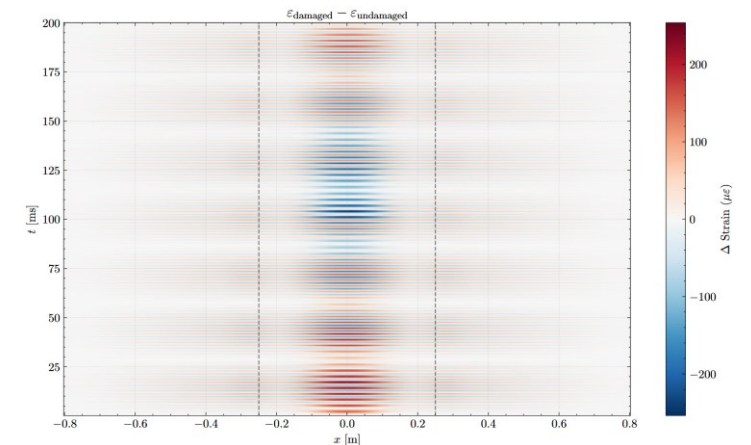
Undamaged



Damaged

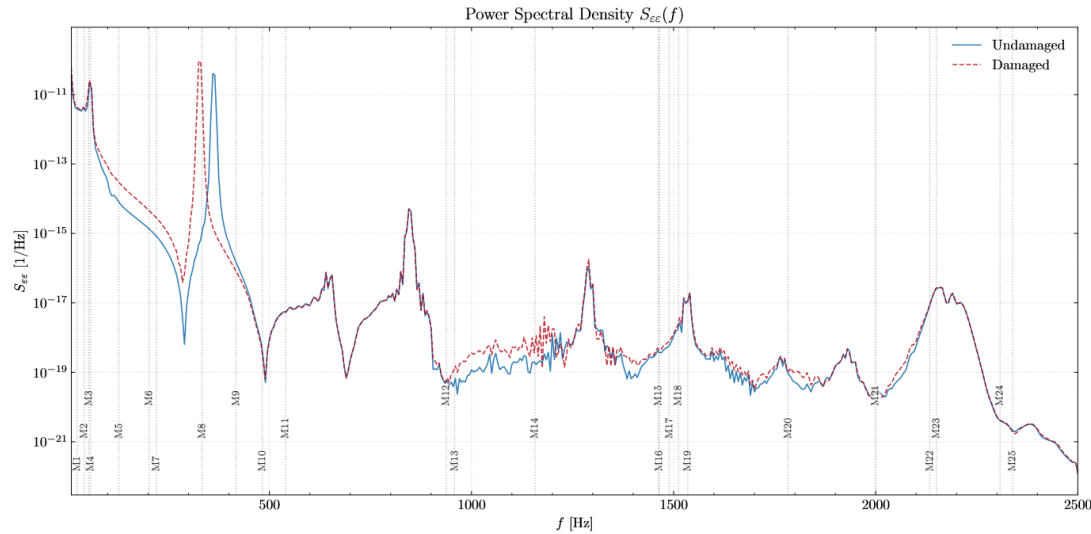


Difference

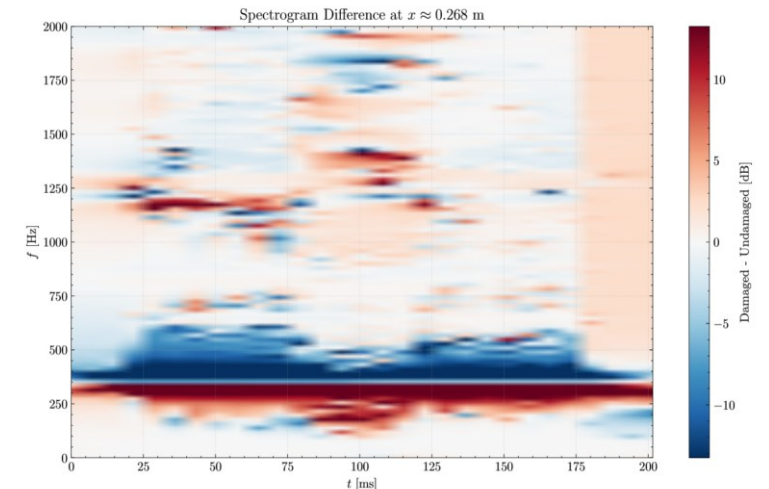
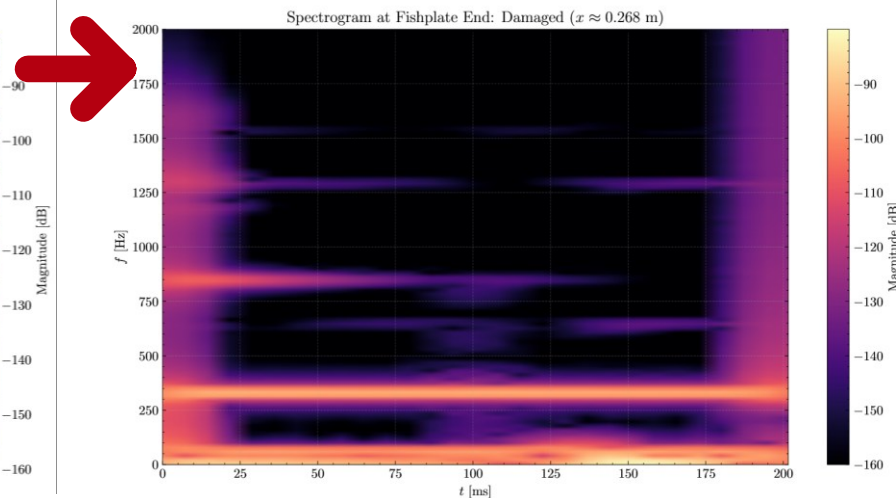
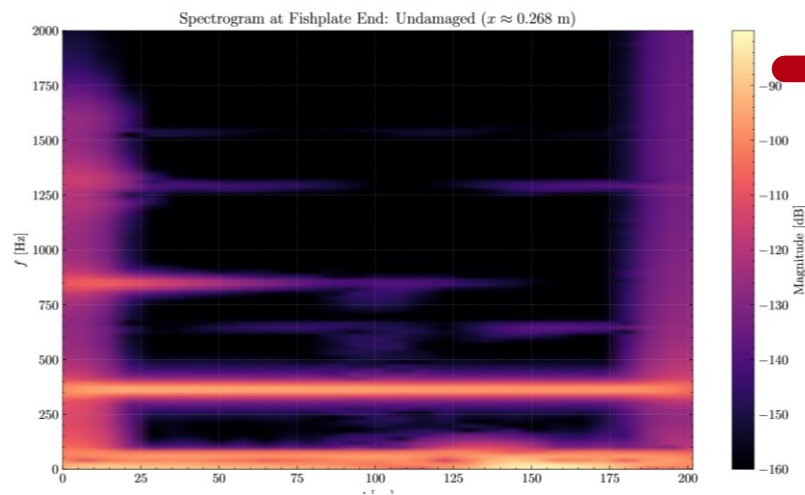


- **Localized** strain variation (**$\pm 200 \mu\epsilon$**)
- Difference **strictly confined** within the **fishplate** boundaries (**± 0.25 m**)
- The damaged joint acts as a compliant connection undergoing **oscillatory opening and closing cycles**

Spectral analysis



- Low frequency semi-rigid modes **unaffected by the damage**
- Manifest frequency shift from **360Hz** to **330Hz** from decreased local stiffness



Mesh size verification

Target: at least 4-6 elements per wavelength in order to resolve the wave

Benchmark frequency: Nyquist frequency limit $f_n = 2500$ Hz, provides a **conservative** safety margin as frequencies of interest are an order of magnitude lower

Wave propagation equations:

- **Longitudinal**

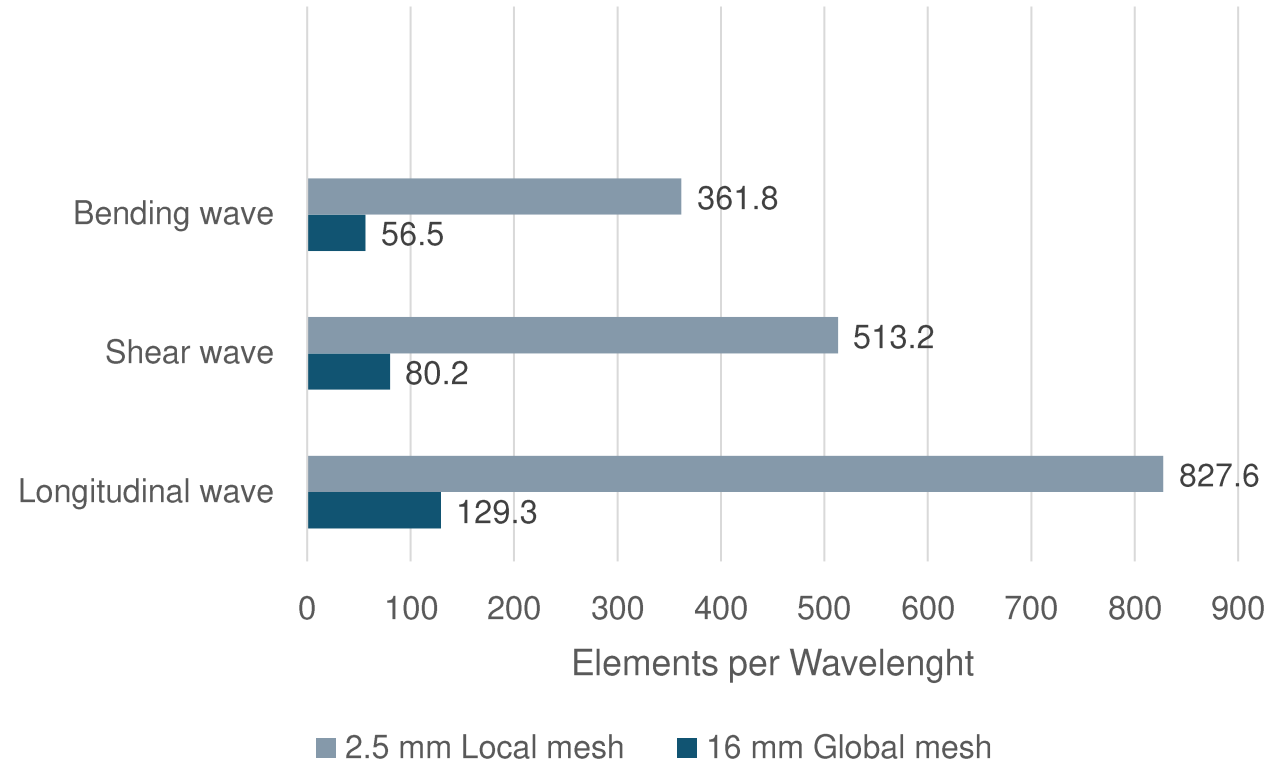
$$\lambda_L = \frac{C_L}{f_N} = \frac{\sqrt{E/\rho}}{f_n} \approx 2069 \text{ mm}$$

- **Shear**

$$\lambda_S = \frac{C_S}{f_N} = \frac{\sqrt{G/\rho}}{f_n} \approx 3206 \text{ mm}$$

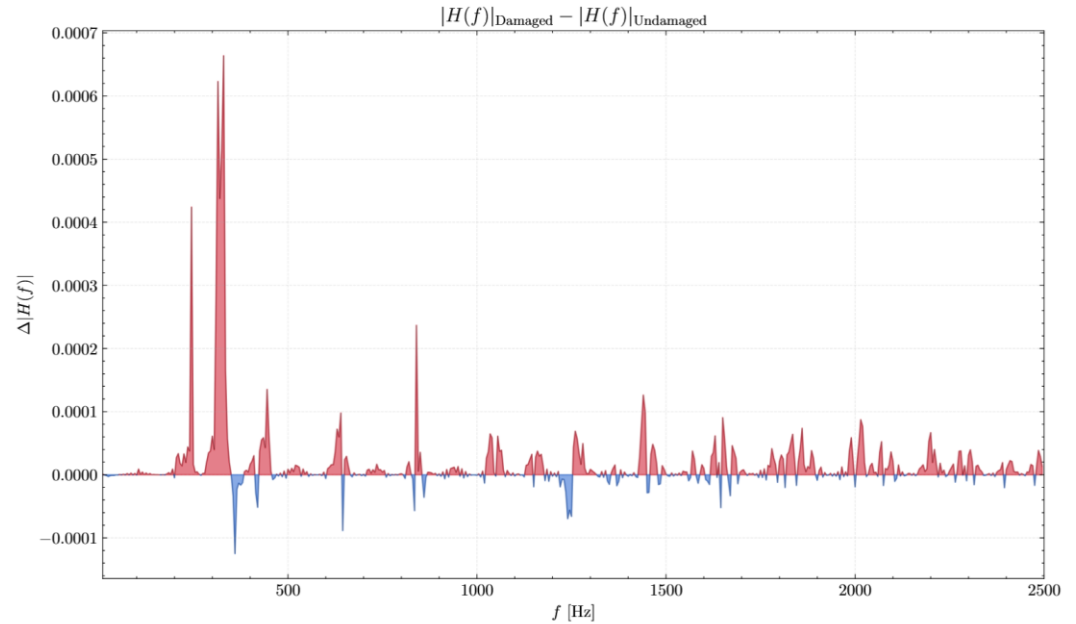
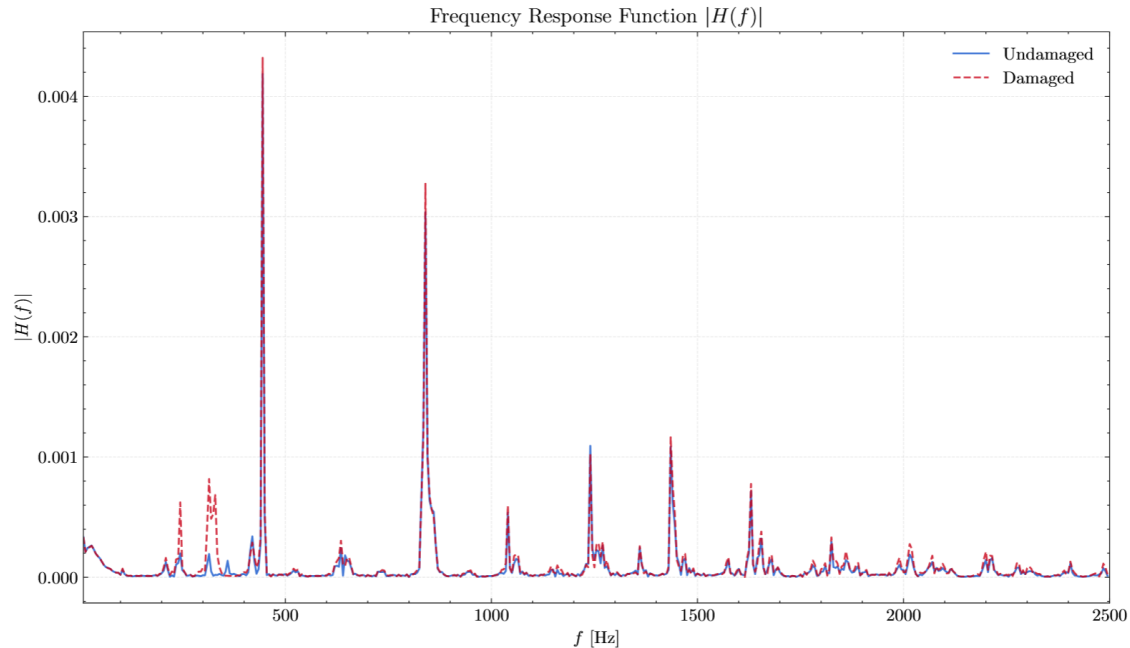
- **Bending**

$$\lambda_B = \sqrt{\frac{2\pi}{f_N} \left(\frac{EI}{\rho A} \right)^{1/4}} \approx 904.6 \text{ mm}$$



Even the worst-case scenario (16mm element on bending waves) **exceeds by an order of magnitude** the minimum number of elements required

Spectral analysis



Frequency response function

- Averaged probe spectrum divided by the 1D hertzian load spectrum
- Removes external force influence
- The frequency shift is still **clearly present**

$$H_i(f) = \frac{\mathcal{F}\{\varepsilon_i(t)\}}{\mathcal{F}\{f_i(t)\}}$$

$$H(f) = \frac{1}{N_{\text{probes}}} \sum_{i=1}^{N_{\text{probes}}} |H_i(f)|$$

Damage detection strategy

The danger:

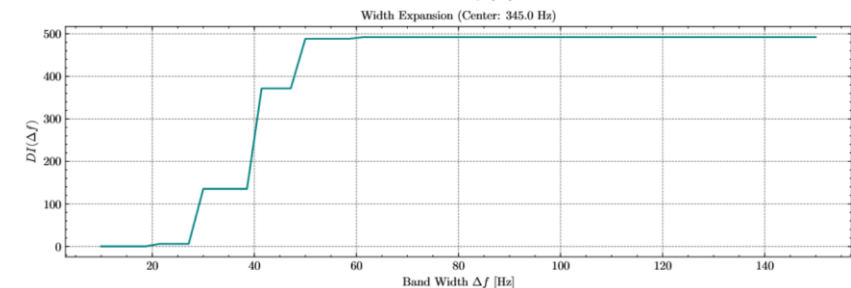
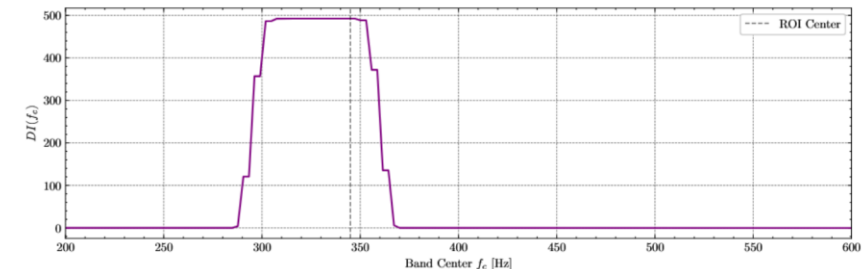
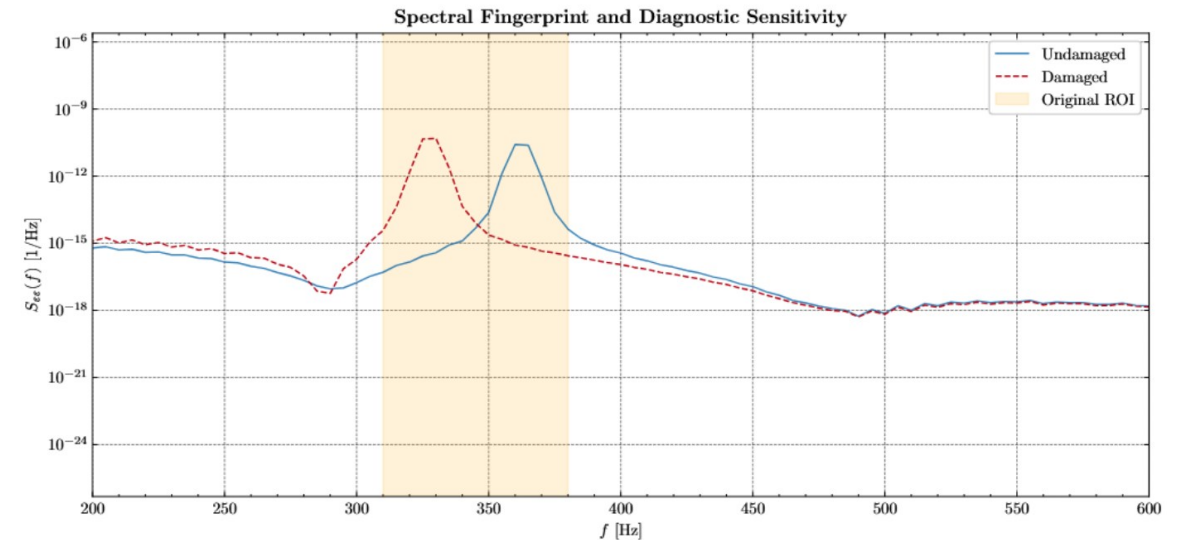
- Broken end-posts cause electrical short circuits leading to **False Clear** (collision risk) or **False Occupancy** (rail network blockage) signals

Diagnostic Index:

- Frequency band: $[f_1, f_2] = [310, 380] \text{ Hz}$
- $DI \gg 1$ **Failure of the end-post**

$$DI = \int_{f_1}^{f_2} \frac{S_{\varepsilon\varepsilon, \text{measured}}(f)}{S_{\varepsilon\varepsilon, \text{undamaged}}(f)} df$$

- Sensitivity analysis** to band centre and bandwidth
- For this simulation the **DI** returns a value of around **500**, **correctly detecting the complete failure of the end-post**.
- A partial failure would **return lower values**





Sant'Anna
Scuola Universitaria Superiore Pisa